

Designed by Remus Teodorescu 2026

LECTURE 10.2

Solid Oxide Cells

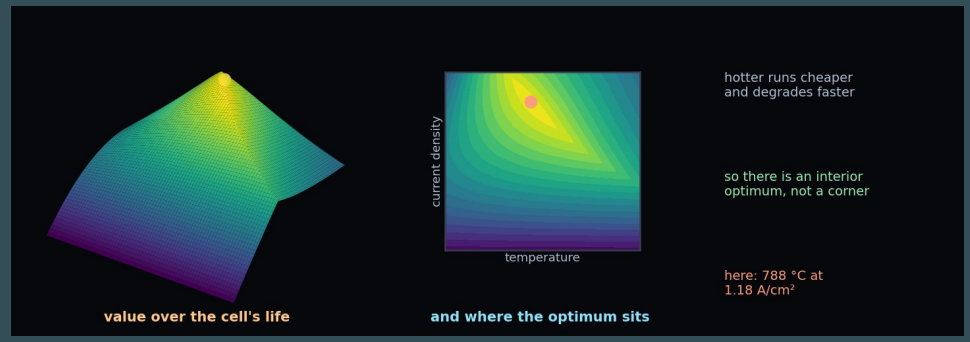
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME



Schematic: the value a cell produces over its lifetime against temperature and current, with its best point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the steam channel, and finds the best voltage by a gradient through one PINN trained over the whole range of voltages.

Learning Objectives

SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam running short, a current that follows the voltage

3. **The best voltage** — one PINN for every voltage, the worth and its gradient

Incropera carries the transport, Liu the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.

READ, AND WHAT EACH ANSWERS

Incropera ch. 1	where do the losses sit?
Incropera ch. 14	how does the steam run short?
Liu ch. 5	what does a gradient buy?

The Solid Oxide Electrolyser

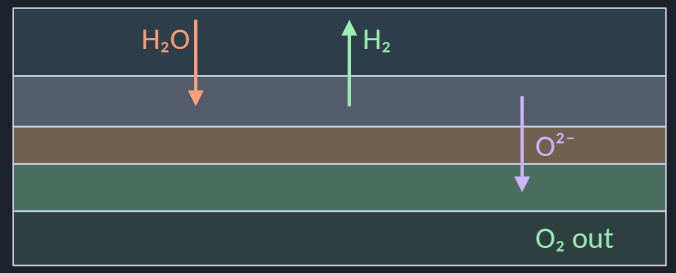
Solid oxide electrolysis cell — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only O^{2-} ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

ΔH the energy to split water, ΔG the part that must be electricity, $T\Delta S$ the part heat can supply

THE CELL, RUN BACKWARDS



steam channel
fuel electrode, porous Ni
electrolyte: O^{2-} only
air electrode
air channel

AGAINST THE LOW-TEMPERATURE CELLS

	alkaline	PEM	solid oxide
ion carried	OH^-	H^+	O^{2-}
temperature	60 – 90 °C	50 – 80 °C	700 – 850 °C
catalysts	nickel	Pt, Ir	nickel, ceramics
kWh per kg H_2	50 – 55	50 – 55	about 40, with steam

Typical values from the literature, rounded. The gain is real only where heat is available: $T\Delta S$ comes from a neighbouring process, so about a quarter less electricity per kilogram of hydrogen.

Three Electrolysers

PEM electrolyser — polymer electrolyte membrane cell, run at 50 to 80 degrees, whose electrolyte is a humid polymer that conducts protons H^+. Liquid water is split on the oxygen side, and the protons cross the membrane and form hydrogen on the other side.

THE PEM CELL, AGAINST THE SOLID OXIDE CELL

ion and direction	H^+ to the hydrogen side, not O^{2-} to the air side
water fed	liquid, to the oxygen side, not steam to the fuel side
hydrogen out	nearly pure and under pressure, not mixed with steam
catalysts	platinum and iridium, not nickel and ceramic oxides
varying load	followed within seconds; no stack kept hot

WHERE EACH FITS IN ELECTRIFICATION

	alkaline	PEM	solid oxide
power it suits	steady supply	wind, solar	steady, with heat
installed today	most capacity	second	first MW plants
cost per kW	lowest	higher	highest
main limit	slower, large	iridium supply	stack lifetime
best fit	bulk hydrogen	wind, solar sites	industry, e-fuels

Typical positions from the literature, rounded. No single winner: alkaline and PEM cells carry the present build-out on wind and solar; the solid oxide cell pays where steady power meets industrial heat, and only if the stack lasts.

The Polarisation Curve

Polarisation curve — the cell voltage against the current density: the Nernst potential of the gas, plus the losses. Each loss has its own range - activation at low current, ohmic in the middle, gas transport at the end - and in electrolysis the losses add to the voltage; in a fuel cell they subtract.

THE NERNST POTENTIAL, AND THE LOSSES ON TOP

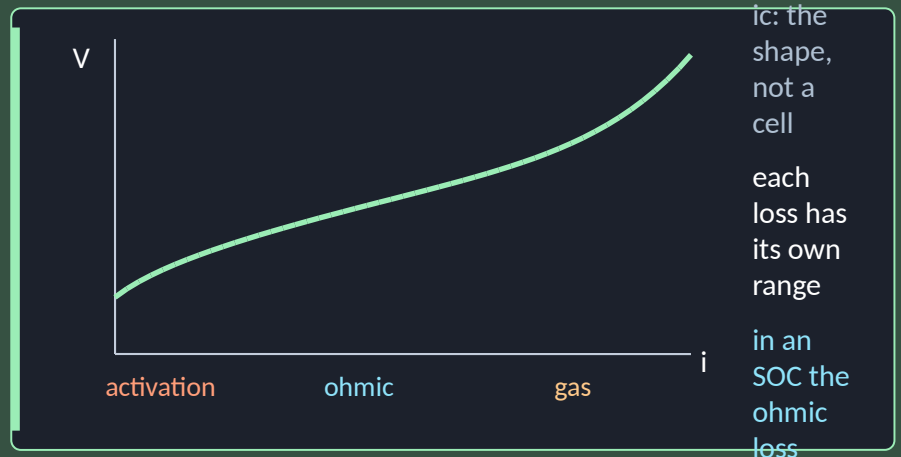
$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

E the Nernst potential, y the steam fraction, p_{O_2} the air side's oxygen, V the cell voltage $\eta_{act}, \eta_{ohm}, \eta_{conc}$ the activation, ohmic and gas-transport losses; in electrolysis they add

The exercise gathers all three losses into one area-specific resistance, 0.25 Ω cm², an estimate that stands for the curve at its operating point.

THE SHAPE OF THE CURVE



schematic: the shape, not a cell

each loss has its own range

in an SOC the ohmic loss leads

in an SOC the ohmic loss leads

why a ceramic's conductivity

Button-cell curves are the validation data.

The Thermoneutral Voltage

Thermoneutral voltage — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero and the temperature stays even. For steam near 800 degrees it is 1.29 volts, and stacks are designed at or just above it.

THE BALANCE, AND THE HEAT IN ONE TERM

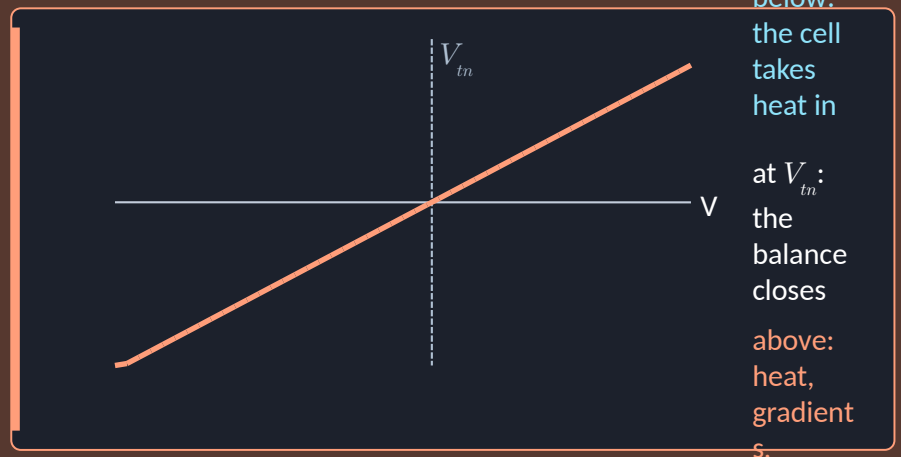
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

V_{tn} the thermoneutral voltage, ΔH the enthalpy of splitting, q the heat per electrode area, i the current density

TALK · 2 MIN How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, what does running just above the dashed line cost, and what does it buy?

THE HEAT AGAINST THE VOLTAGE



below: the cell takes heat in

at V_{tn}: the balance closes

above: heat, gradients, ageing

one clean term all the losses gather in q

the catch ASR falls as T rises

A hotter spot draws more current, and heats further.

The Steam Channel

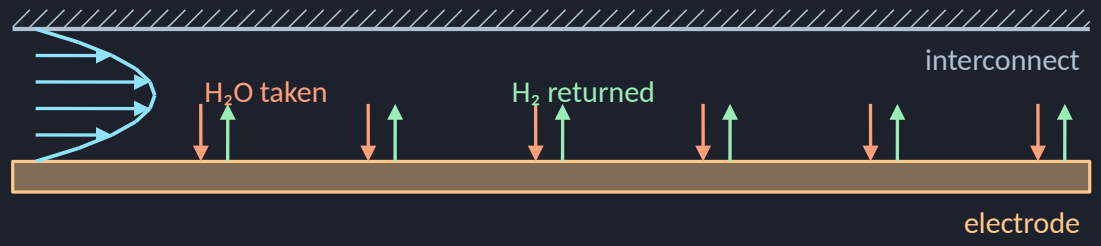
Steam channel — the fuel channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s, the electrode takes steam and returns hydrogen, and the steam fraction falls along the channel. The flow is the same known Poiseuille; what changes is the source, which is no longer a given flux.

THE SCALED EQUATION OF EX_10.2

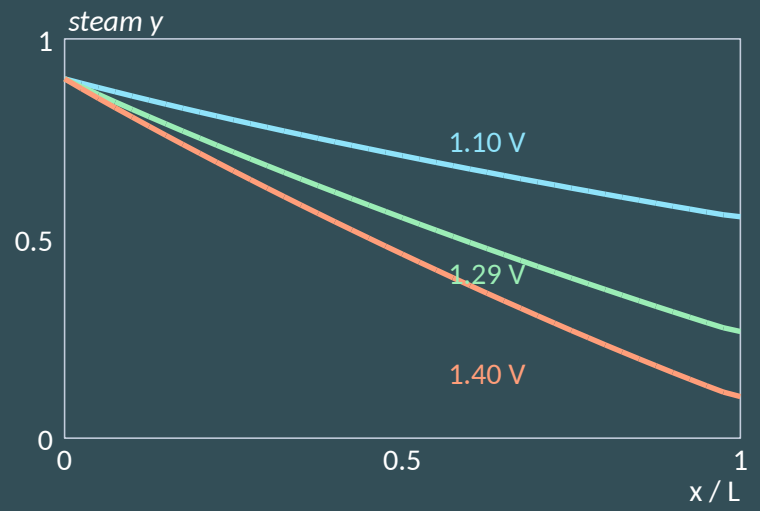
$$y_x = \frac{y_{xx}}{Pe} - K(V - E(y)), \qquad Pe = 125, \quad K = 1.83\ \mathrm{V}^{-1}$$

y the steam fraction, x along the channel over L, Pe the Péclet number, K gathers L, u, h, c, F, ASR

STEAM TAKEN, HYDROGEN RETURNED



STEAM ALONG THE CHANNEL



The exercise's reference at three cell voltages: at 1.29 V the outlet is 0.27 - 69 % of the steam used. Too far, and the outlet starves.

The Local Current

Current from the voltage — the line that changes everything from the last lecture: the wall flux is no longer given. The cell sits at one voltage, the local gas sets the local Nernst potential, and their gap over the resistance is the local current - highest where the steam is fresh.

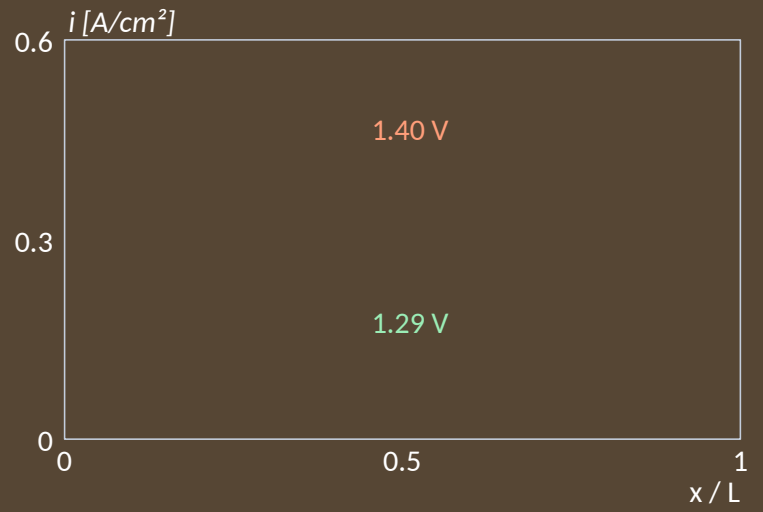
ONE VOLTAGE, A CURRENT THAT VARIES

$$i(y) = \frac{V - E(y)}{ASR}$$

i the local current density, $E(y)$ the Nernst potential of the local gas, ASR all losses as one area resistance (an estimate)

So the equation is nonlinear in y through $E(y)$, and the outlet needs a condition again: the exercise holds the slope at zero there with a loss term, and solves the reference with a boundary-value solver rather than a march.

THE CURRENT ALONG THE CHANNEL



Highest at the inlet, where the steam is fresh. A starved outlet runs a higher local loss and ages faster: the cell fails at one end first.

One PINN for Every Voltage

Parametric PINN — the move of the hydrogen channel again: the cell voltage joins the position as an input, and after one training over 1.10 to 1.40 V any voltage is read off, not solved again. The inlet is built into the trial function; the zero-slope outlet is a loss term.

THE TRIAL FUNCTION, AND ITS RESIDUAL

$$\hat{y}(x, V) = y_{in} + x\,\mathcal{N}(x, V)$$

$y_{in} = 0.90$ the inlet, built in, $\mathcal{N}$ the network: x and the cell voltage V in, trained over 1.10 to 1.40 V

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)      # x and the cell voltage V in
y = Y_IN + x * net(p)                           # the trial function: the inlet
built in
g = grad(y, p)                                  # first derivatives by autograd
res = g[:,0:1] - d2(y, p, 0) / PE + K * (V - nernst(y))   # the residual
out = grad(y, p)[outlet, 0:1]                   # the outlet slope, held at 0
```

THREE METHODS AT 1.29 V

	FVM, 50 cells	PINN
worst error	0.0042	0.0005
hydrogen	5.20 g/h	5.21 g/h
one solution	2 ms	under 1 ms
a new voltage	solve again	read off
training, offline	none	20 s

The reference is a boundary-value solver at tolerance 10^{-10}. The same network is within 0.0013 of it at 1.10, 1.20, 1.35 and 1.40 V: one training, the whole range.

The Worth of an Hour

Objective — what the operation is for, written as one number: the hydrogen made in an hour, at its price, less the electricity used, at its price. The physics supplies the two rates; the prices are assumptions, not physics, and the result is conditional on them.

THE OBJECTIVE, AND ITS GRADIENT

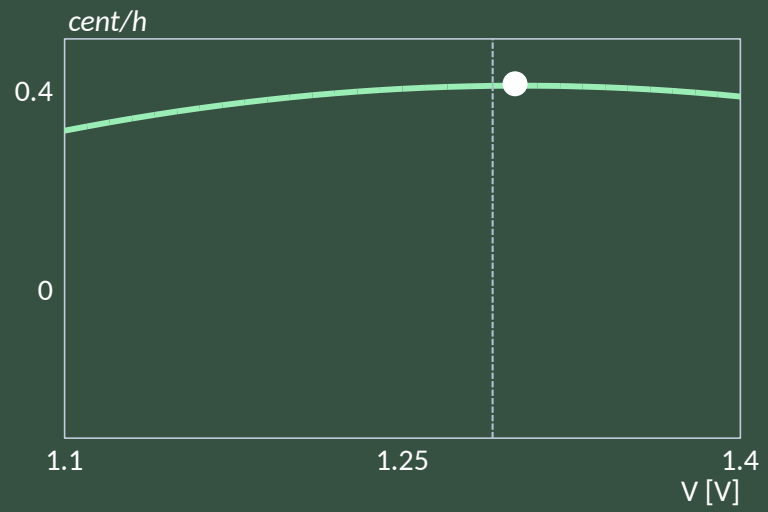
$$J(V) = p_{H_2}\,\dot{m}_{H_2}(V) - p_{el}\,P(V), \qquad \frac{dJ}{dV} \text{ by autograd}$$

J the worth of an hour, p_{H_2}, p_{el} the prices (assumptions, not physics), $\dot{m}_{H_2}$ hydrogen made, P power drawn

THE BEST VOLTAGE IN PYTORCH

```
V = torch.tensor([1.25], requires_grad=True)   # the knob, differentiable
for k in range(200):
    J = value_per_hour(net, V)                 # hydrogen minus electricity
    V = V + lr * grad(J, V)                    # uphill, through the physics
```

CENT PER HOUR AGAINST VOLTAGE



by the gradient	1.30 V, 200 steps, 0.2 s
by 61 solver runs	1.30 V, 4 s

No Best Point Over a Lifetime

Degradation — the slow rise of the cell's resistance: nickel coarsens, interfaces delaminate, chromium poisons the air electrode. The rate is Arrhenius in T and a power of i, so what produces more now ages faster - and published rates span an order of magnitude.

THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

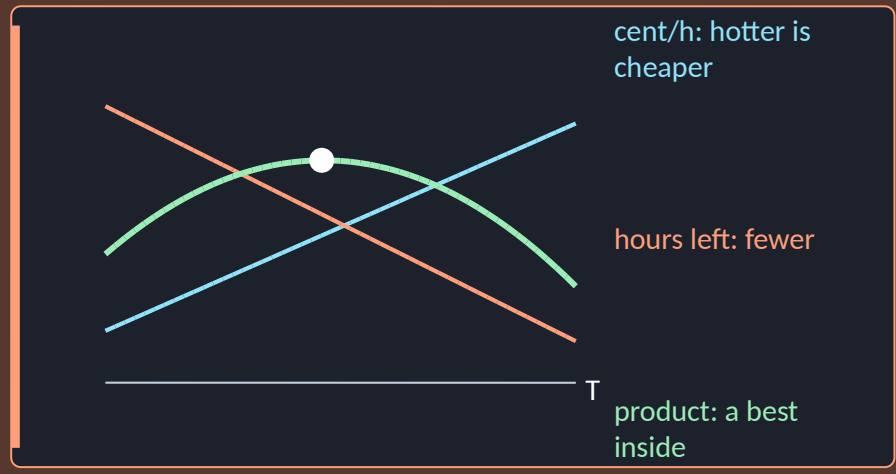
$$\frac{d\,ASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\max_{V(t),\,T(t)} \int_0^{t_{life}} [p_{H_2}\dot{m}_{H_2} - p_{el}P]\,dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

k_0, E_d, n the degradation law's constants : Arrhenius in T, a power of i; published rates span an order of magnitude

TALK · 2 MIN How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

THE TRADE, DRAWN



so the answer a trajectory $V(t), T(t)$

held back by lifetime, and $|\nabla T|$: ceramic

The Gradient Through the Physics

Differentiable model — a model that gives the fields and their derivatives with respect to anything, by the autograd that trained it. A solver evaluates the physics; optimising through it costs a solve per variable per step, or an adjoint. Through a PINN the gradient is read off.

WHAT EACH PREDICTED FIELD SUPPLIES

the field	from	what it supplies here
temperature	L8	the $\lvert\nabla T\rvert$ limit the ceramic sets
composition	L9	the local Nernst potential
current, potential	L10	the degradation it drives

Part 2's argument in one table: none of those fields was an end in itself. They are inputs to a decision, and constraints are written in terms of them.

SOLVER AGAINST NETWORK, OPTIMISING

solver	one solve per variable per step
or	an adjoint, built by hand
network	the gradient from autograd
in Ex_10.2	0.2 s against 4 s of sweeps
to refuse	a lifetime as a promise

Degradation rates are uncertain by an order of magnitude: a predicted lifetime ranks schedules, nothing more.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the channel with its temperature	the cell cools below V_{tn}; current follows	Incropera ch. 5	stacks; MP10.2A
the degradation law from records	k_0, E_d, n fitted to long voltage drifts	Ex_10.2 truth	lifetime models; MP10.2B
the button-cell fit	kinetics and resistance against measured curves	Incropera ch. 1	validation before a channel
the daily schedule	the hourly electricity price as the driver	Liu ch. 5	grid-coupled operation; L12
the steam supply and its heat	exchangers that make the gain real	Incropera ch. 11	plant efficiency

WHY THEY ARE LEFT OUT

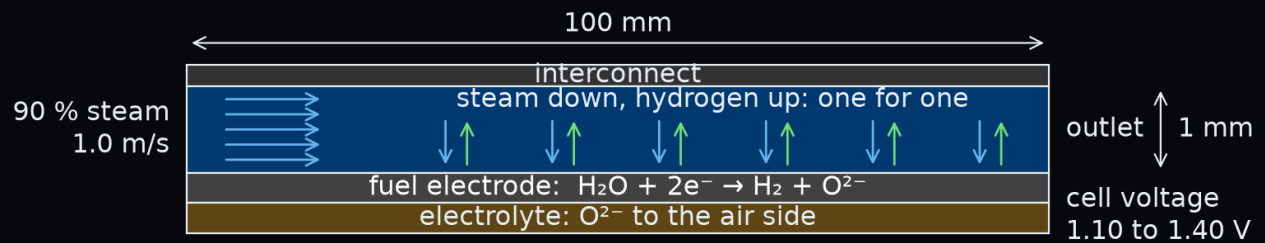
Ex_10.2 solves one isothermal channel at one moment, scored against a boundary-value solver, so none of them is needed to do it. The first two rows are the two mini projects; the last two are where the schedule meets the grid of L12.

EXERCISE

Exercise 10.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its channel, the Nernst potential and K
- **4** — the boundary-value reference; finite volumes with Newton
- **5** — one PINN for every voltage from 1.10 to 1.40 V
- **6** — the best cell voltage, by a gradient through the network
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B



THE QUESTION EACH PART ANSWERS

1 – 3	what sets the local current?
4	how fine must the grid be?
5	one network, every voltage?
6	which voltage pays best?

REPORT QUESTION, TO CONCLUDE

The ASR, the diffusivity and the prices are the notebook's estimates, and two students with different price ratios should find different best voltages. What in your answer is physics, and what is an assumption?

SUMMARY

Key Takeaways

SOLID OXIDE CELL

what it is	a ceramic cell, two directions
function	steam to hydrogen, heat helping
train, tune	run at or just above V_{tn}
vs low temperature	a quarter less electricity
used for	storing surplus power as fuel

LIFETIME OPTIMISATION

what it is	a PINN with the voltage as input
function	the worth, and its gradient
train, tune	once, over 1.10 to 1.40 V
vs a solver	the gradient comes free
used for	a trajectory, not a setting

Next — L11.1: a network in a real-time loop, where there is no PDE at all.

Questions

1 – 2 · A NETWORK FOR THE STEAM CHANNEL

1 **How can a network learn the steam fraction along the channel of the cell for every cell voltage?**

The network takes the position along the channel and the cell voltage, and is trained over 1.10 to 1.40 V until the species equation holds at points along the channel, with the local current set by the gap between the cell voltage and the local Nernst potential. The steam fraction at the inlet is built into the form of the solution, and the zero slope at the outlet is held by a loss term. After one training, any voltage in the range is read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

2 **Why does the local current make the steam channel harder to solve, and how is the network's answer checked?**

The flux at the electrode is no longer given: the current at each point follows the local steam fraction, so the equation is nonlinear in it and needs a condition at the outlet, and it can no longer be marched from the inlet. The network takes the nonlinear term into its residual with no change of method. Its answer is scored against a boundary-value solver at a tight tolerance: at 1.29 V within 0.0005 of the steam fraction, where finite volumes on 50 cells are within 0.0042. *Ex_10.2 sections 3 – 5*

3 – 4 · THE BEST VOLTAGE

3 **How can the best cell voltage be found with a network, and what does the gradient save?**

The worth of an hour, the hydrogen made at its price less the electricity used at its price, is computed from the network's steam fraction and current. With the voltage as an input, automatic differentiation gives the derivative of the worth with respect to it, and gradient ascent finds 1.30 V in 200 steps and 0.2 s; a sweep of 61 runs of the reference solver finds the same voltage in 4 s. The prices are assumptions, so the voltage found is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

4 **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of the voltage and the temperature, because what produces more now ages the cell faster. A solver gives the worth, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives the gradient with respect to every input by automatic differentiation. Degradation rates are uncertain by an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

NEXT

L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

BEFORE THE NEXT SESSION

read	Liu ch. 5
run	Ex_10.2, sections 1 to 6
bring	your best voltage, and its worth
think about	what was physics, what was price

Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.